

Disproof of conjecture 00000009939: entropy minus the longest bar's contribution can exceed $\frac{1}{2} \log(\#\text{bars})$

Jackmeson1

2026-10-05

1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: Persistence entropy: the Shannon functional of persistence bar length distributions. Conjecture: The persistence entropy satisfies concentration inequalities and is mutually bounded with the Shannon entropy of the longest bar: the entropy minus longest-bar contribution is bounded by half the logarithm of the bar count; the constant of the bound is tight.

Chinese text (translated). The Chinese version says that the persistence entropy is mutually bounded with the Shannon entropy of the longest bar, and makes this precise as: “the entropy minus the longest bar’s contribution is bounded by half of the logarithm of the number of bars; the constant of the bound is tight.”

Reading. The conjecture is a conjunction. Its explicit quantitative conjunct is

$$(B) \quad E(B) - c_B(I) \leq \frac{1}{2} \log n,$$

for every nonempty barcode B with n bars and every longest bar I of B , where E is the persistence entropy and $c_B(I)$ is the contribution of I to E , i.e. its own summand $-p_I \log p_I$. We refute (B), which refutes the conjunction. We do not address the vague clause “satisfies concentration inequalities”. The tightness clause presupposes (B); we also show that no constant $c < 1$ can replace $\frac{1}{2}$ (Theorem 5).

Conventions. Natural logarithm. A change of base multiplies both sides of (B) by the same positive factor, so the result does not depend on the base. $\eta(x) = -x \log x$ (Mathlib `Real.negMulLog`). Bars are counted with multiplicity; in all our witnesses the bars are pairwise distinct intervals, so counting distinct bars gives the same n . All witnesses have a *unique* longest bar, so the choice of “the longest bar” is not an issue.

2 Definitions (as formalized)

Definition 1. A *bar* is a finite interval $[b, d)$ with real $b < d$; its length is $\ell = d - b > 0$. A *barcode* is a finite multiset of bars. For a barcode B with lengths $\ell_1, \dots, \ell_n$, put $L = \sum_i \ell_i$ and $p_i = \ell_i/L$. The *persistence entropy* is

$$E(B) = - \sum_{i=1}^n p_i \log p_i = \sum_{i=1}^n \eta(\ell_i/L),$$

and the *contribution* of a bar I of length ℓ_I is $c_B(I) = \eta(\ell_I/L)$. A bar $I \in B$ is a *longest bar* if $\ell_J \leq \ell_I$ for all $J \in B$.

This is the Shannon entropy of the distribution (p_i) given by the normalized bar lengths, which is the standard notion. Atienza, Gonzalez-Diaz and Rucco (arXiv:1701.07857) write: “Persistent entropy is defined as the Shannon entropy of the persistence barcode of a given filtration.” Atienza, Gonzalez-Diaz and Soriano-Trigueros (arXiv:1803.08304) note: “Observe that, to compute persistent entropy, we only have to consider the length of each interval”, and “in the definition of persistent entropy, we assume that there are no infinite length intervals in the persistence barcode.” This is why bars are finite here.

Realization by a filtration. Every finite multiset of finite bars $[b_j, d_j]$, $j = 1, \dots, k$, is the degree-1 persistence barcode of a filtered simplicial complex. Take k hollow triangles T_j that share one vertex v . Let v enter at time $\min_j b_j$, let the other two vertices and the three edges of T_j enter at time b_j , and let the 2-simplex filling T_j enter at time d_j . At time t , H_1 is freely generated by the classes $[\partial T_j]$ with $b_j \leq t < d_j$, and each inclusion map sends $[\partial T_j]$ to itself (or to 0 once T_j is filled). Hence H_1 is the direct sum of the interval modules on $[b_j, d_j]$. This construction is prose only; Mathlib has no persistent homology, so the Lean input is the barcode itself.

3 The witnesses

For $a > 0$ and $m \in \mathbb{N}$ let $S(a, m)$ be the barcode consisting of $[0, a]$ and the m unit bars $[i, i + 1]$, $0 \leq i < m$ (Lean `spike a ha m`). It has $n = m + 1$ bars and total length $L = a + m$.

Lemma 2 (`entropy_spike`, `longest_eq`, `gap_spike`, `spike_nodup`). $E(S(a, m)) = \eta(a/L) + m\eta(1/L)$. If $a \neq 1$ the n bars are pairwise distinct. If $a > 1$ then $[0, a]$ is the unique longest bar and

$$E(S(a, m)) - c([0, a]) = m\eta(1/L) = \frac{m}{a + m} \log(a + m).$$

Proof. The contribution of $[0, a]$ is $\eta(a/L)$ and each unit bar contributes $\eta(1/L)$, so the formula for E follows. A unit bar has length $1 < a$, so it is not longest. Finally $\eta(1/L) = -(1/L) \log(1/L) = (\log L)/L$. $\square$

Theorem 3 (`bound_fails_every_count`). For every $n \geq 2$, the barcode $S(2, n - 1)$ has exactly n pairwise distinct bars and a unique longest bar I , and $E - c(I) = \frac{n-1}{n+1} \log(n + 1) > \frac{1}{2} \log n$.

Proof. Write $m = n - 1 \geq 1$ (Lean `key_ineq`). If $m = 1$, the claim is $\frac{1}{3} \log 3 > \frac{1}{2} \log 2$, i.e. $\log 9 > \log 8$. If $m \geq 2$, then $\frac{m}{m+2} \geq \frac{1}{2}$ and $\log(m + 2) > \log(m + 1) \geq 0$, so $\frac{m}{m+2} \log(m + 2) \geq \frac{1}{2} \log(m + 2) > \frac{1}{2} \log(m + 1)$. $\square$

The smallest example with three bars (`witness_211`). The bars $[0, 2]$, $[0, 1]$, $[1, 2]$ have lengths $(2, 1, 1)$, $L = 4$ and $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. Then $E = \frac{3}{2} \log 2$, the longest bar contributes $\frac{1}{2} \log 2$, and $E - c = \log 2 > \frac{1}{2} \log 3$, because $4 > 3$.

Theorem 4 (`conjecture9939_false`). Statement (B) is false.

Proof. Apply Theorem 3 with $n = 3$. $\square$

Theorem 5 (`no_constant_below_one`). For every real $c < 1$ there is a barcode, with a longest bar, such that $E - c(I) > c \log n$ for every longest bar I .

(The Lean statement concludes exactly this. The witness used is the barcode `spike 2` plus unit bars, whose longest bar is unique by the separately proved lemma `longest_eq`; uniqueness is therefore established in the development, though not stated in this theorem’s conclusion.)

Proof. Choose $m \in \mathbb{N}$ with $m > 2/(1 - c)$. Then $m(1 - c) > 2 > 2c$, so $\frac{m}{m+2} > c$. Since $\log(m+2) > \log(m+1) > 0$, we get $\frac{m}{m+2} \log(m+2) > \frac{m}{m+2} \log(m+1) > c \log(m+1)$. $\square$

4 Alternative readings of “the Shannon entropy of the longest bar”

The Chinese text fixes the reading “contribution” = $c(I)$. Two other readings of the English phrase are also refuted, by the barcode $S(16, 48)$ (one bar $[0, 16)$ and 48 unit bars, $n = 49$, $L = 64$, $p_{\max} = \frac{1}{4}$).

- *Binary entropy* $h(p_{\max}) = \eta(p_{\max}) + \eta(1 - p_{\max})$ (Mathlib `Real.binEntropy`). Lemma `binGap_spike` gives $E - h(p_{\max}) = \frac{m}{L} \log m = \frac{3}{4} \log 48$, and $\frac{3}{4} \log 48 > \frac{1}{2} \log 49$ because $48^3 > 49^2$ (`binary_reading_false`).
- *Self-information / min-entropy* $-\log p_{\max}$. Lemma `minGap_spike` gives $E + \log p_{\max} = \frac{m}{L} \log a = \frac{3}{4} \log 16 = \log 8$, and $\log 8 > \log 7 = \frac{1}{2} \log 49$ (`minEntropy_reading_false`).

Since all gaps above are positive, the bound also fails if “bounded by” is read as $|E - c(I)| \leq \frac{1}{2} \log n$.

Remark 6 (not formalized). If “bar length distribution” meant instead the empirical distribution of the length *values* (a histogram), the bars $[0, 1)$, $[0, 2)$, $[0, 3)$ would give a uniform histogram on three values, with entropy $\log 3$ and longest-bar contribution $\frac{1}{3} \log 3$. The difference $\frac{2}{3} \log 3$ again exceeds $\frac{1}{2} \log 3$. This reading is non-standard and is not part of the Lean development.

5 Lean formalization and axiom audit

File `lean/Conjecture9939/Basic.lean` (296 lines, only `import Mathlib`, Lean toolchain and `Mathlib` as in `lake-manifest.json`). Main definitions: `Bar`, `Bar.length`, `Barcode := Multiset Bar`, `totalLength`, `contribution`, `persistenceEntropy`, `barCount`, `IsLongestBar`, `ConjecturedBound`, `BinaryBound` and `MinEntropyBound`. Main theorem:

`theorem conjecture9939_false :  $\neg$ ConjecturedBound`

where `ConjecturedBound` is

$\forall B : \text{Barcode}, B \neq 0 \rightarrow \forall I : \text{Bar}, \text{IsLongestBar } B \ I \rightarrow$
 $\text{persistenceEntropy } B - \text{contribution } B \ I \leq \frac{1}{2} \log(\text{barCount } B).$

The other theorems are `bound_fails_every_count`, `witness_211`, `no_constant_below_one`, `binary_reading_false` and `minEntropy_reading_false`. For each of them, `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`. `lake build` succeeds.